

Binomial Distribution

Key Notes and Formulae

1. When is X binomial? (check “BINS”)

- **Binary** – each trial has only two outcomes: success or failure.
- **Independent** – the trials do not affect each other (e.g. *with replacement*).
- **Number of trials** n is fixed.
- **Same probability of success** p on every trial.

Then we write $X \sim B(n, p)$, where X = number of successes in n trials.

2. Formulae (in the IB formula booklet)

Probability of exactly r successes	$P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}, \quad r = 0, 1, \dots, n$
Mean (expected value)	$E(X) = np$
Variance	$\text{Var}(X) = np(1 - p)$
Standard deviation	$\sigma = \sqrt{np(1 - p)}$

3. Using the GDC

- $P(X = r) \rightarrow \text{binomPdf}(n, p, r)$ (one value)
- $P(a \leq X \leq b) \rightarrow \text{binomCdf}(n, p, a, b)$ (a range – always *inclusive*)

Always write down the distribution, e.g. “ $X \sim B(20, 0.25)$ ”, and the probability statement before the answer.

4. Translating the words (X takes whole-number values only)

Words	Inequality	GDC ($n = 20$)
at least 4	$X \geq 4$	$\text{binomCdf}(20, p, 4, 20)$
more than 4	$X > 4 \iff X \geq 5$	$\text{binomCdf}(20, p, 5, 20)$
at most 4	$X \leq 4$	$\text{binomCdf}(20, p, 0, 4)$
fewer / less than 4	$X < 4 \iff X \leq 3$	$\text{binomCdf}(20, p, 0, 3)$
at least one	$X \geq 1$	$1 - P(X = 0)$

5. Conditional probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad \text{e.g.} \quad P(X \leq 4 | X \geq 2) = \frac{P(2 \leq X \leq 4)}{P(X \geq 2)}$$

Tip: the top is the *overlap* of the two conditions; the bottom is the condition after “given that”.

6. “The k th success happens on the n th trial”

This means: exactly $k - 1$ successes in the first $n - 1$ trials, **and** a success on trial n .

$$P = \underbrace{P(Y = k - 1)}_{Y \sim B(n-1, p)} \times p$$

e.g. third prize on the tenth game: $P(T = 2) \times p$, with $T \sim B(9, p)$.

7. “Find the least n ” questions

$$P(\text{at least one success}) > 0.95 \iff 1 - (1 - p)^n > 0.95 \iff (1 - p)^n < 0.05.$$

Use the GDC table (or logs) and take the **smallest whole number** n that works.

8. “Within one standard deviation of the mean”

$P(\mu - \sigma < X < \mu + \sigma)$: work out the end values, then round **inwards** to whole numbers.

e.g. $P(230.2 < T < 249.8) = P(231 \leq T \leq 249)$.

9. Discrete probability distributions (useful with binomial questions)

All probabilities add up to 1	$\sum P(X = x) = 1$
Expected value	$E(X) = \sum x P(X = x)$
Combined events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Complementary events	$P(A') = 1 - P(A)$

10. Worked example

A fair die is rolled 10 times. X = number of sixes, so $X \sim B\left(10, \frac{1}{6}\right)$.

- $E(X) = 10 \times \frac{1}{6} = \frac{5}{3} \approx 1.67$, $\text{Var}(X) = 10 \times \frac{1}{6} \times \frac{5}{6} = \frac{25}{18} \approx 1.39$
- $P(X = 2) = \binom{10}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^8 \approx 0.291$
- $P(X \geq 1) = 1 - \left(\frac{5}{6}\right)^{10} \approx 0.838$

How to use this worksheet: questions go from easy to difficult. Attempt each page first, then check the yellow answer page that follows it.

Question 1 Calculator Easy

[Maximum mark: 6]

A multiple choice exam consists of 20 questions. Each question has four answers. Only one of the answers is correct. For each question, Jack randomly chooses one of the four answers.

- (a) Find the expected number of questions Jack answers correctly. [2]
- (b) Find the probability that Jack:
- (i) answers exactly four questions correctly;
 - (ii) answers more than four questions correctly. [4]

Working:

Question 2 Calculator Medium

[Maximum mark: 8]

A factory makes toys. The probability that a toy is defective is 0.08. A random sample of 50 toys is tested.

- (a) Find the expected number of defective toys. [2]
- (b) Find the probability that there is at least one defective toy in the sample. [3]
- (c) Given that there is at least one defective toy in the sample, find the probability that there are at most three defective toys. [3]

Working:

Answers – Questions 1 & 2

Question 1

Let X be the number of questions Jack answers correctly. $X \sim B\left(20, \frac{1}{4}\right)$.

(a) $E(X) = np = 20 \times \frac{1}{4} = \boxed{5}$

(b) (i) $P(X = 4) = \text{binomPdf}\left(20, \frac{1}{4}, 4\right) \approx \boxed{0.190}$

(ii) $P(X > 4) = P(X \geq 5) = \text{binomCdf}\left(20, \frac{1}{4}, 5, 20\right) \approx \boxed{0.585}$

Question 2

Let X be the number of defective toys in the sample. $X \sim B(50, 0.08)$.

(a) $E(X) = np = 50 \times 0.08 = \boxed{4 \text{ defective toys}}$

(b) $P(X \geq 1) = 1 - P(X = 0) = \text{binomCdf}(50, 0.08, 1, 50) \approx \boxed{0.985}$

(c) $P(X \leq 3 \mid X \geq 1) = \frac{P(1 \leq X \leq 3)}{P(X \geq 1)} = \frac{\text{binomCdf}(50, 0.08, 1, 3)}{P(X \geq 1)} \approx \frac{0.410}{0.985} \approx \boxed{0.416}$

Question 3 Calculator Medium

[Maximum mark: 8]

AB Technologies produces DC motors of which 5% are defective. The motors are packaged into boxes, each box containing 60 motors. A box is selected at random.

- (a) Find the probability that the box contains:
- (i) exactly one defective motor;
 - (ii) at least two defective motors. [4]
- (b) Given that there are at least two defective motors in the box, find the probability that there are at most four defective motors. [2]

James orders 5 boxes of motors for his company.

- (c) Find the probability that James will receive at least one but no more than fifteen defective motors in his order. [2]

Working:

Question 4 Calculator Medium

[Maximum mark: 6]

The probability of obtaining tails when a biased coin is tossed is 0.6. The coin is tossed 400 times.

- (a) (i) Write down the mean number of tails.
- (ii) Find the variance of the number of tails. [3]
- (b) Find the probability that the number of tails obtained is less than one standard deviation away from the mean. [3]

Working:

Answers – Questions 3 & 4

Question 3

(a) Let X be the number of defective motors in the box. $X \sim B(60, 0.05)$.

(i) $P(X = 1) = \text{binomPdf}(60, 0.05, 1) \approx \boxed{0.145}$

(ii) $P(X \geq 2) = \text{binomCdf}(60, 0.05, 2, 60) \approx \boxed{0.808}$

(b) $P(X \leq 4 \mid X \geq 2) = \frac{P(2 \leq X \leq 4)}{P(X \geq 2)} = \frac{\text{binomCdf}(60, 0.05, 2, 4)}{P(X \geq 2)} \approx \frac{0.628}{0.808} \approx \boxed{0.777}$

(c) Let Y be the number of defective motors in 5 boxes ($5 \times 60 = 300$ motors). $Y \sim B(300, 0.05)$.

$$P(1 \leq Y \leq 15) = \text{binomCdf}(300, 0.05, 1, 15) \approx \boxed{0.568}$$

Question 4

Let T be the number of tails in 400 tosses. $T \sim B(400, 0.6)$.

(a) (i) $E(T) = np = 400 \times 0.6 = \boxed{240}$

(ii) $\text{Var}(T) = np(1 - p) = 400 \times 0.6 \times 0.4 = \boxed{96}$

(b) $\sigma = \sqrt{96} \approx 9.80$

$$P(|T - 240| < 9.80) = P(230.2 < T < 249.8) = P(231 \leq T \leq 249)$$

$$= \text{binomCdf}(400, 0.6, 231, 249) \approx \boxed{0.668}$$

Question 5 Calculator Medium

[Maximum mark: 6]

The probability of obtaining heads when a biased coin is tossed is 0.65. The coin is tossed seven times. Find the probability of obtaining:

(a) at least four heads; [2]

(b) the fourth tail is on the seventh toss. [4]

Working:

Question 6 Calculator Hard

[Maximum mark: 6]

A bag contains 8 blue marbles, 12 green marbles and m red marbles. A marble is selected at random and replaced. This is performed three times.

(a) Write down the probability that the first marble selected is blue. [1]

Let X be the number of blue marbles selected.

(b) Find the smallest value of m for which $\text{Var}(X) < 0.5$. [5]

Working:

Answers – Questions 5 & 6

Question 5

- (a) Let H be the number of heads in seven tosses. $H \sim B(7, 0.65)$.

$$P(H \geq 4) = \text{binomCdf}(7, 0.65, 4, 7) \approx \boxed{0.800}$$

- (b) The 4th tail on the 7th toss means exactly 3 tails in the first 6 tosses, then a tail on toss 7.

Let T be the number of tails in the first six tosses. $T \sim B(6, 0.35)$.

$$P(T = 3) = \text{binomPdf}(6, 0.35, 3) \approx 0.23549$$

By independence of the tosses:

$$P(\text{4th tail on 7th toss}) = P(T = 3) \times P(\text{tail on 7th toss}) = 0.23549 \times 0.35 \approx \boxed{0.0824}$$

Question 6

(a) $P(\text{blue}) = \boxed{\frac{8}{20+m}}$

(b) $X \sim B\left(3, \frac{8}{20+m}\right)$, so

$$\text{Var}(X) = np(1-p) = 3 \left[\frac{8}{20+m} \right] \left[1 - \frac{8}{20+m} \right] = \frac{24(12+m)}{(20+m)^2}$$

Solve $\text{Var}(X) < 0.5$ for $m > 0$:

$$\frac{24(12+m)}{(20+m)^2} < \frac{1}{2} \Rightarrow 48(12+m) < (20+m)^2 \Rightarrow m > 17.9 \quad [\text{by GDC}]$$

Hence the smallest value of $m \in \mathbb{Z}^+$ is $\boxed{18}$.

Question 7 Calculator Hard

[Maximum mark: 15]

A burger shop serves veggie burgers and chicken burgers. Each customer will choose one of the following three options; buy only a veggie burger, buy only a chicken burger or buy both a veggie burger and a chicken burger. The probability that a customer buys a veggie burger is 0.65 and the probability that a customer buys a chicken burger is 0.54.

(a) Find the probability that a customer chosen at random will buy:

(i) both a veggie burger and a chicken burger;

(ii) only a chicken burger.

[5]

On a typical day, 600 customers come to the burger shop.

(b) Find:

(i) the expected number of veggie burgers sold on a typical day;

(ii) the probability that more than 400 veggie burgers will be sold on a typical day.

[4]

It is known that 44% of the customers who come to the burger shop are females, and that 75% of the female customers buy a chicken burger.

(c) (i) A customer is selected at random. Find the probability that the customer is female and buys a chicken burger.

(ii) A male customer is selected at random. Find the probability that he buys a chicken burger.

[6]

Working:

Answers – Question 7

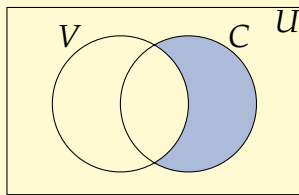
Question 7

Let V = buys a veggie burger, C = buys a chicken burger. Every customer buys at least one, so $P(V \cup C) = 1$.

(a) (i) $P(V \cup C) = P(V) + P(C) - P(V \cap C)$

$$1 = 0.65 + 0.54 - P(V \cap C) \Rightarrow \boxed{P(V \cap C) = 0.19}$$

(ii) Only chicken is the shaded region $V' \cap C$:



$$\begin{aligned} P(V' \cap C) &= P(C) - P(V \cap C) \\ &= 0.54 - 0.19 = \boxed{0.35} \end{aligned}$$

(b) Let X be the number of veggie burgers sold. $X \sim B(600, 0.65)$.

(i) $E(X) = 600 \times 0.65 = \boxed{390 \text{ veggie burgers}}$

(ii) $P(X > 400) = \text{binomCdf}(600, 0.65, 401, 600) \approx \boxed{0.185}$

(c) Let F = female, M = male.

(i) $P(C | F) = \frac{P(C \cap F)}{P(F)} \Rightarrow 0.75 = \frac{P(C \cap F)}{0.44} \Rightarrow \boxed{P(C \cap F) = 0.33}$

(ii) $P(C) = P(C \cap F) + P(C \cap M) \Rightarrow 0.54 = 0.33 + P(C \cap M) \Rightarrow P(C \cap M) = 0.21$

$P(M) = 1 - 0.44 = 0.56$

$$P(C | M) = \frac{P(C \cap M)}{P(M)} = \frac{0.21}{0.56} = \boxed{0.375}$$

Question 8 Calculator Hard

[Maximum mark: 15]

A random variable X has a probability distribution given in the following table.

x	0	1	2
$P(X = x)$	p	$\frac{8}{15}$	q

- (a) Given that $E(X) = 0.8$, find the exact value of: (i) q ; (ii) p . [4]

A bag contains black and red balls, with at least two of each colour. Two balls are drawn from the bag without replacement. The number of red balls drawn is given by the random variable X .

- (b) (i) Write down the probability of drawing two red balls.
(ii) Explain why the probability of drawing two black balls is $\frac{1}{3}$.
(iii) The bag contains a total of ten balls of which k are black. Find k . [5]

A game is played in which two balls are drawn from the bag of ten balls, without replacement. A player wins a prize if two black balls are drawn.

- (c) Olivia plays the game twelve times. Find the probability that she wins exactly three prizes. [2]
(d) William plays the game until he wins three prizes. Find the probability that he wins his third prize on his tenth attempt. [4]

Working:

Answers – Question 8

Question 8

(a) (i) $E(X) = \sum xP(X = x)$:

$$0.8 = 0 \times p + 1 \times \frac{8}{15} + 2q \Rightarrow 2q = \frac{8}{10} - \frac{8}{15} = \frac{24}{30} - \frac{16}{30} = \frac{8}{30} \Rightarrow \boxed{q = \frac{2}{15}}$$

$$(ii) \sum P(X = x) = 1: \quad p + \frac{8}{15} + \frac{2}{15} = 1 \Rightarrow p + \frac{2}{3} = 1 \Rightarrow \boxed{p = \frac{1}{3}}$$

$$(b) (i) P(X = 2) = q = \boxed{\frac{2}{15}}$$

(ii) Two black balls means zero red balls: $P(\text{two black}) = P(X = 0) = p = \frac{1}{3}$.

$$(iii) \frac{k}{10} \times \frac{k-1}{9} = \frac{1}{3} \Rightarrow k^2 - k = 30 \Rightarrow (k-6)(k+5) = 0 \Rightarrow \boxed{k = 6} \text{ (since } k > 0)$$

(c) Let S be the number of prizes in 12 games. $S \sim B\left(12, \frac{1}{3}\right)$.

$$P(S = 3) = \text{binomPdf}\left(12, \frac{1}{3}, 3\right) \approx \boxed{0.212}$$

(d) The 3rd prize on the 10th attempt means exactly 2 prizes in the first 9 games, then a win on game 10.

Let T be the number of prizes in 9 games. $T \sim B\left(9, \frac{1}{3}\right)$.

$$P(T = 2) = \text{binomPdf}\left(9, \frac{1}{3}, 2\right) \approx 0.23411$$

$$\begin{aligned} P(\text{3rd prize on 10th game}) &= P(T = 2) \times P(\text{win 10th game}) \\ &= 0.23411 \times \frac{1}{3} \approx \boxed{0.0780} \end{aligned}$$

Question 9 Calculator Hard

[Maximum mark: 21]

Ten balls numbered 2, 2, 3, 3, 3, 5, 5, 5, 5, 5 are placed in a bag. Balls are taken one at a time from the bag at random and the number noted.

Throughout the question a ball is always replaced before the new ball is taken.

- (a) A single ball is taken from the bag. Let X denote the value shown on the ball. Find $E[X]$. [2]
- (b) Three balls are taken from the bag. Find the probability that:
 - (i) the sum of the three numbers is 9;
 - (ii) the median of the three numbers is 3. [6]
- (c) Twelve balls are taken from the bag. Find the probability that more than five of the balls are numbered 2. [3]
- (d) Determine the least number of balls that must be taken from the bag for the probability of taking out at least one ball numbered 5 to be greater than 0.9. [3]
- (e) Another bag also contains balls numbered 2, 3 or 5. Nine balls are to be taken from this bag at random. It is calculated that the expected number of balls numbered 2 is 4.5, and the variance of the number of balls numbered 3 is 1.25. Find the least possible number of balls numbered 5 in this bag. [7]

Working:

Answers – Question 9

Question 9

(a)

x	2	3	5
$P(X = x)$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{5}{10}$

$$E[X] = 2 \times \frac{2}{10} + 3 \times \frac{3}{10} + 5 \times \frac{5}{10} = \boxed{3.8}$$

(b) (i) Sum 9: $\{2, 2, 5\}$ in any order (3 ways) or $\{3, 3, 3\}$:

$$3 \left[\frac{2}{10} \times \frac{2}{10} \times \frac{5}{10} \right] + \frac{3}{10} \times \frac{3}{10} \times \frac{3}{10} = 0.06 + 0.027 = \boxed{0.087}$$

(ii) Median 3: $\{2, 3, 3\}$ (3 ways), $\{3, 3, 3\}$, $\{3, 3, 5\}$ (3 ways), $\{2, 3, 5\}$ (6 ways):

$$3(0.2)(0.3)(0.3) + (0.3)^3 + 3(0.3)(0.3)(0.5) + 6(0.2)(0.3)(0.5) \\ = 0.054 + 0.027 + 0.135 + 0.18 = \boxed{0.396}$$

(c) Let Y be the number of balls numbered 2. $Y \sim B(12, 0.2)$.

$$P(Y > 5) = \text{binomCdf}(12, 0.2, 6, 12) \approx \boxed{0.0194}$$

(d) Let Z be the number of balls numbered 5 among N balls. $Z \sim B(N, 0.5)$.

$$1 - P(Z = 0) > 0.9 \Rightarrow (0.5)^N < 0.1 \Rightarrow N = 4, 5, 6, \dots \text{ [by GDC]}$$

(check: $0.5^3 = 0.125$, $0.5^4 = 0.0625$)

The least number of balls is $\boxed{4}$.

(e) Let $p = P(\text{ball 2})$ and $q = P(\text{ball 3})$. $S \sim B(9, p)$, $T \sim B(9, q)$.

$$E[S] = 9p = 4.5 \Rightarrow p = \frac{1}{2}. \quad \text{Var}(T) = 9q(1 - q) = 1.25 \Rightarrow q = \frac{1}{6} \text{ or } q = \frac{5}{6}.$$

$q = \frac{5}{6}$ is rejected since $\frac{1}{2} + \frac{5}{6} > 1$. So $q = \frac{1}{6}$.

$P(5) = 1 - \frac{1}{2} - \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$. The smallest bag is 6 balls (3 twos, 1 three, 2 fives).

Least possible number of balls numbered 5 = $\boxed{2}$

Question 10 Calculator Hard

[Maximum mark: 22]

Ten balls numbered 1, 2, 2, 3, 3, 3, 4, 4, 4, 4 are placed in a bag. Balls are taken one at a time from the bag at random and the number noted.

Throughout the question a ball is always replaced before the new ball is taken.

- (a) A single ball is taken from the bag. Let X denote the value shown on the ball. Find $E[X]$. [2]
- (b) Three balls are taken from the bag. Find the probability that:
 - (i) the sum of the three numbers is 8;
 - (ii) the median of the three numbers is 4. [6]
- (c) Fifteen balls are taken from the bag. Find the probability that less than six of the balls are numbered 3. [3]
- (d) Find the least number of balls that must be taken from the bag for the probability of taking out at least one ball numbered 4 to be greater than 0.95. [3]
- (e) Another bag also contains balls numbered 1, 2, 3 or 4. Twenty balls are to be taken from this bag at random. It is calculated that the expected number of balls numbered 2 is 2.5, the expected number of balls numbered 3 is 5, and the variance of the number of balls numbered 1 is 1.8. Find the least possible number of balls numbered 4 in this bag. [8]

Working:

Answers – Question 10

Question 10

(a) $E[X] = 1 \times \frac{1}{10} + 2 \times \frac{2}{10} + 3 \times \frac{3}{10} + 4 \times \frac{4}{10} = \frac{30}{10} = \boxed{3}$

(b) (i) Sum 8: $\{1, 3, 4\}$ (6 ways), $\{2, 2, 4\}$ (3 ways), $\{2, 3, 3\}$ (3 ways):

$$6(0.1)(0.3)(0.4) + 3(0.2)(0.2)(0.4) + 3(0.2)(0.3)(0.3) \\ = 0.072 + 0.048 + 0.054 = \boxed{0.174}$$

(ii) Median 4 needs at least two 4s: $3P(144) + 3P(244) + 3P(344) + P(444)$

$$= 3 \left[\frac{1}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \right] + 3 \left[\frac{2}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \right] + 3 \left[\frac{3}{10} \cdot \frac{4}{10} \cdot \frac{4}{10} \right] + \left(\frac{4}{10} \right)^3 = \boxed{0.352}$$

(c) Let Y be the number of balls numbered 3. $Y \sim B(15, 0.3)$.

$$P(Y < 6) = \text{binomCdf}(15, 0.3, 0, 5) \approx \boxed{0.722}$$

(d) Let Z be the number of balls numbered 4 among N balls. $Z \sim B(N, 0.4)$.

$$1 - P(Z = 0) > 0.95 \Rightarrow (0.6)^N < 0.05 \Rightarrow N = 6, 7, 8, \dots \text{ [by GDC]}$$

(check: $0.6^5 \approx 0.078$, $0.6^6 \approx 0.047$)

The least number of balls is $\boxed{6}$.

(e) Let p, q, r be the probabilities of taking out balls numbered 1, 2 and 3.

$$E[T] = 20q = 2.5 \Rightarrow q = \frac{1}{8}. \quad E[U] = 20r = 5 \Rightarrow r = \frac{1}{4}.$$

$$\text{Var}(S) = 20p(1-p) = 1.8 \Rightarrow p = \frac{1}{10} \text{ or } p = \frac{9}{10} \text{ [by GDC].}$$

$p = \frac{9}{10}$ is rejected since $\frac{9}{10} + \frac{1}{8} + \frac{1}{4} > 1$. So $p = \frac{1}{10}$.

$$P(1, 2 \text{ or } 3) = \frac{1}{10} + \frac{1}{8} + \frac{1}{4} = \frac{19}{40} \Rightarrow P(4) = 1 - \frac{19}{40} = \frac{21}{40}$$

The smallest total number of balls in the bag is 40, so the least possible number of balls numbered 4 is $\boxed{21}$.

End of worksheet

Question 11

Calculator

Hard

[Maximum mark: 15]

Jane commutes to work five days a week and travels by either bike or train. When it rains, the probability Jane rides her bike is 0.2. When it does not rain, the probability she catches the train is 0.3. The probability it rains on any given day is 0.4.

- (a) On a randomly selected day, find the probability Jane catches the train to work. [4]
- (b) Given that Jane catches the train, find the probability it was raining that day. [3]
- (c) In a randomly chosen work week, find the probability Jane catches the train on exactly two out of the five days. [3]
- (d) If Jane travels to work 48 weeks a year, riding her bike is free and the train costs \$1.70 per day, calculate how much Jane would expect to pay per year on her commute to work. [5]

Working:

Answers – Question 11

Question 11

Tree diagram: Rain (0.4): bike 0.2, train 0.8. No rain (0.6): train 0.3, bike 0.7.

(a) $P(T) = 0.4 \times 0.8 + 0.6 \times 0.3 = 0.32 + 0.18 = \boxed{0.5}$

(b) $P(R | T) = \frac{P(R \cap T)}{P(T)} = \frac{0.32}{0.5} = \boxed{0.64}$

(c) Let X be the number of days she takes the train. $X \sim B(5, 0.5)$.

$$P(X = 2) = \binom{5}{2}(0.5)^2(0.5)^3 = \frac{10}{32} = \boxed{0.3125}$$

(d) Days per year = $48 \times 5 = 240$. Expected train days = $240 \times 0.5 = 120$.

Expected cost = $120 \times \$1.70 = \boxed{\$204}$